Investigating heterotypic nanoparticle formats for broad antibody neutralization across infections by Zika and dengue viruses

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ABSTRACT

Zika virus (ZV) and dengue virus (DV) are mosquito-borne flaviviruses that cause severe, fatal disease. Four billion people are at risk for infection by DV's four serotypes (DV1-4). Severe ZV and DV cases are due to antibody-dependent enhancement (ADE) interactions between these infections. Currently, there is a need for vaccines that do not carry the risk of inducing ADE. Neutralizing antibodies target the lateral ridge (LR) motif of domain III (DIII) of E glycoprotein, which is critical for viral entry and cell attachment. We previously demonstrated strong immunoprotecting potential of a homotypic ZV immunization using nanoparticles decorated with LR-focused DIII mutant. Here, we studied neutralizing properties of various heterotypic immunizations in mice and found two leading strategies. We analyzed the B cell response to these immunizations via FACS and aim to characterize resulting monoclonal antibodies. This study will further our understanding of the adaptive immune response and vaccine development against flavivirus infections.

BACKGROUND

- DV1-4 and ZV infections transmitted by *Aedes* mosquitoes. Increased prevalence due to climate change
- Symptoms:
  - Mild: flu-like, rash, arthralgia
  - Hemorrhagic fever and shock syndrome (DV), Guillain-Barre syndrome in adults and neonatal microcephaly (ZV)
  - ADE affects prognosis (Fig. 1)

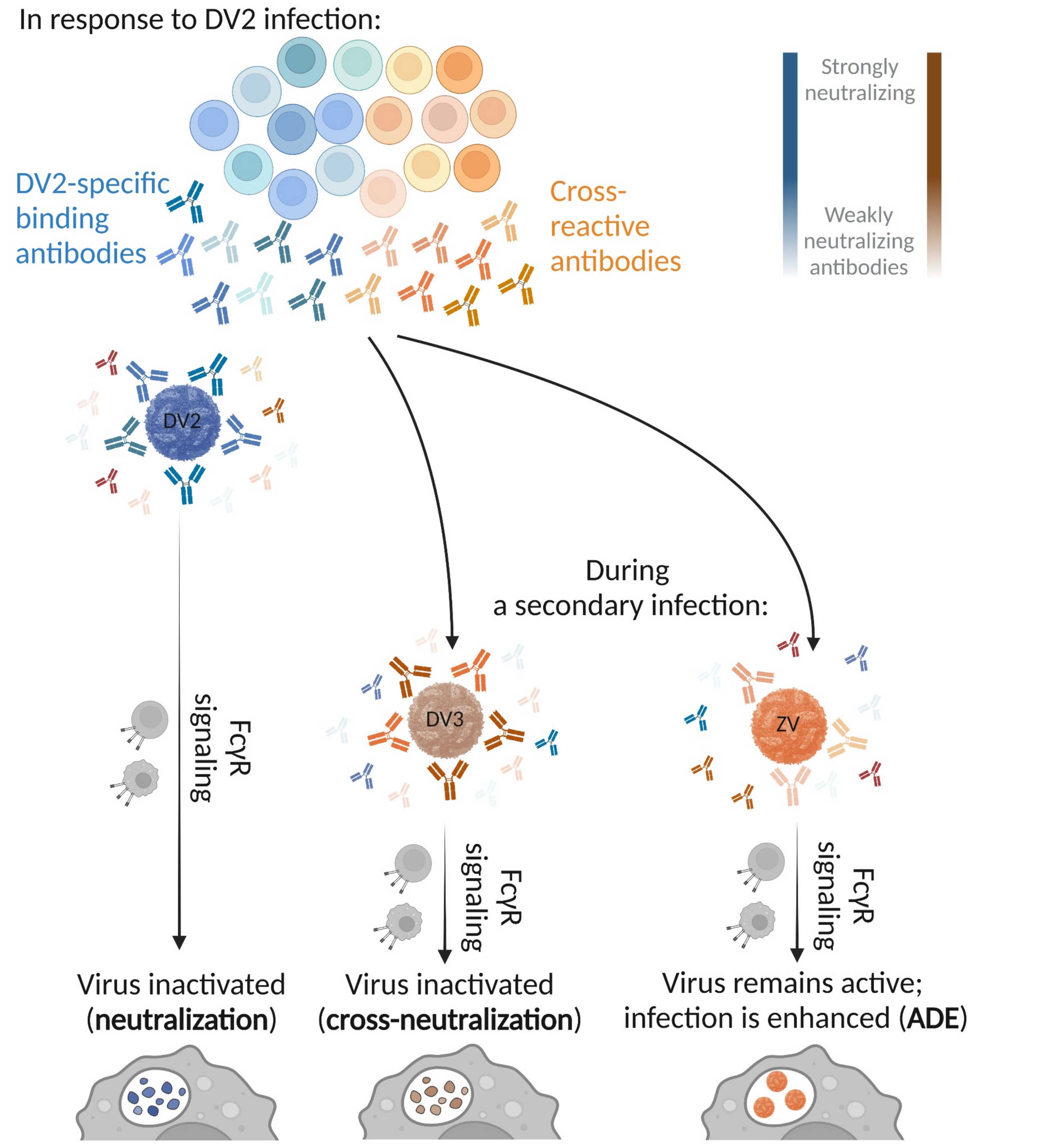
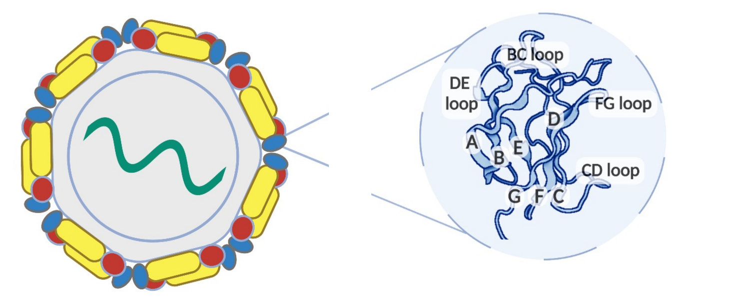


Figure 1. Proposed mechanism of antibody-dependent enhancement (ADE), with DV2 infection as an example of primary infection. Phenomenon occurs due to antibody interactions between infections. During secondary infection, antibodies from primary infection may facilitate viral activity.

- E glycoprotein: cell attachment, viral entry (Fig. 2)
  - Lateral Ridge (LR): targeted by known broadly neutralizing antibodies (bNAbs) across ZV and DV1

BACKGROUND (cont'd)

Figure 2. Flavivirus structural proteins. [L] Proteins on the surface of mature flavivirus virions. [R] Domain III (DIII) of E glycoprotein and its motifs. Lateral Ridge has A-strand, BC-loop, DE-loop, FG-loop.



- B cell response to the LR (Fig. 3): *resurfaced mutants (rsZDIIIs)* on Lumazine Synthase (LS) for multivalent display

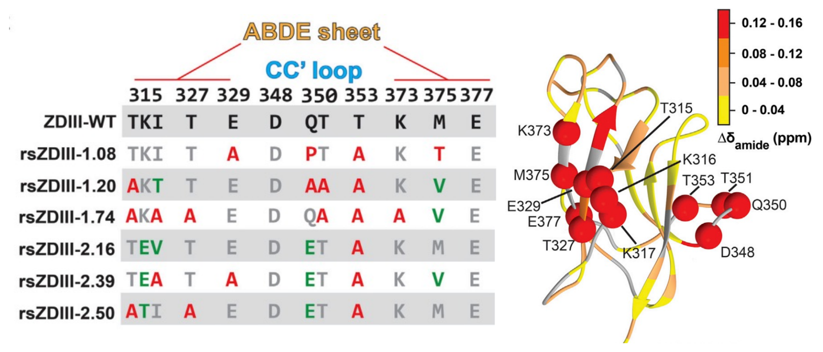


Figure 3. Masked epitopes of rsZVDIIIs.[1] [L] Shortlist of candidates. [R] Chemical shift differences of rsZDIII-2.39 on structural model of wtZDIII (PDB: 5KVG).

- Immunized mice with homotypic nanoparticles (NPs)
  - Sera of *rsZDIII-2.39* demonstrated neutralizing titers against ZV but lacked cross-neutralization activity

- **Objective: ZV-DV immunization to generate high bNAb titers with minimal ADE and to study elicited bNAbs**

RESULTS

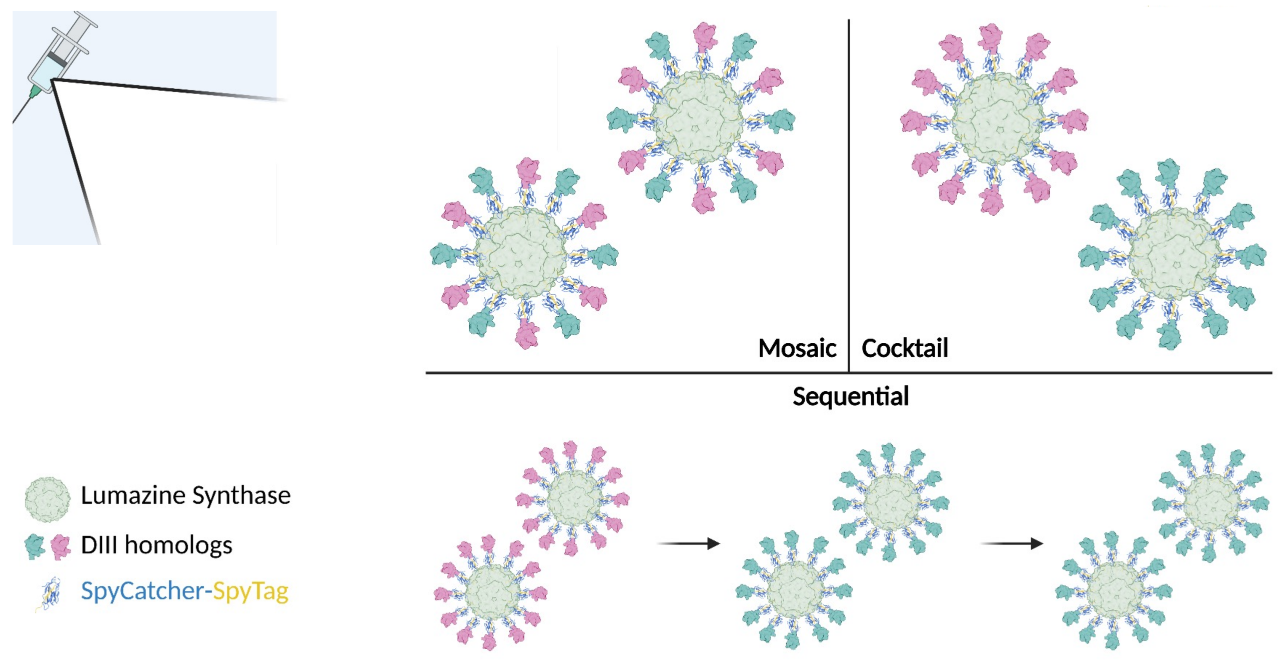


Figure 4. Designs of hetero-NP delivery. *Mosaic* design involves conjugating both DIII homologs onto each NP; *cocktail* as a mixture of homo-NPs; and *sequential* as homo-NPs varied by injection. LS-DIII conjugation via Spy System (PDB: 4MLI), a spontaneous, irreversible reaction. All components recombinantly expressed and purified via affinity resin, anion-exchange chromatography, and size exclusion chromatography.

- rsZDIII-2.39 and wtD1DIII as immunogens (Fig. 4)
- Characterized NP assemblies, e.g. BLI (Fig. 5)

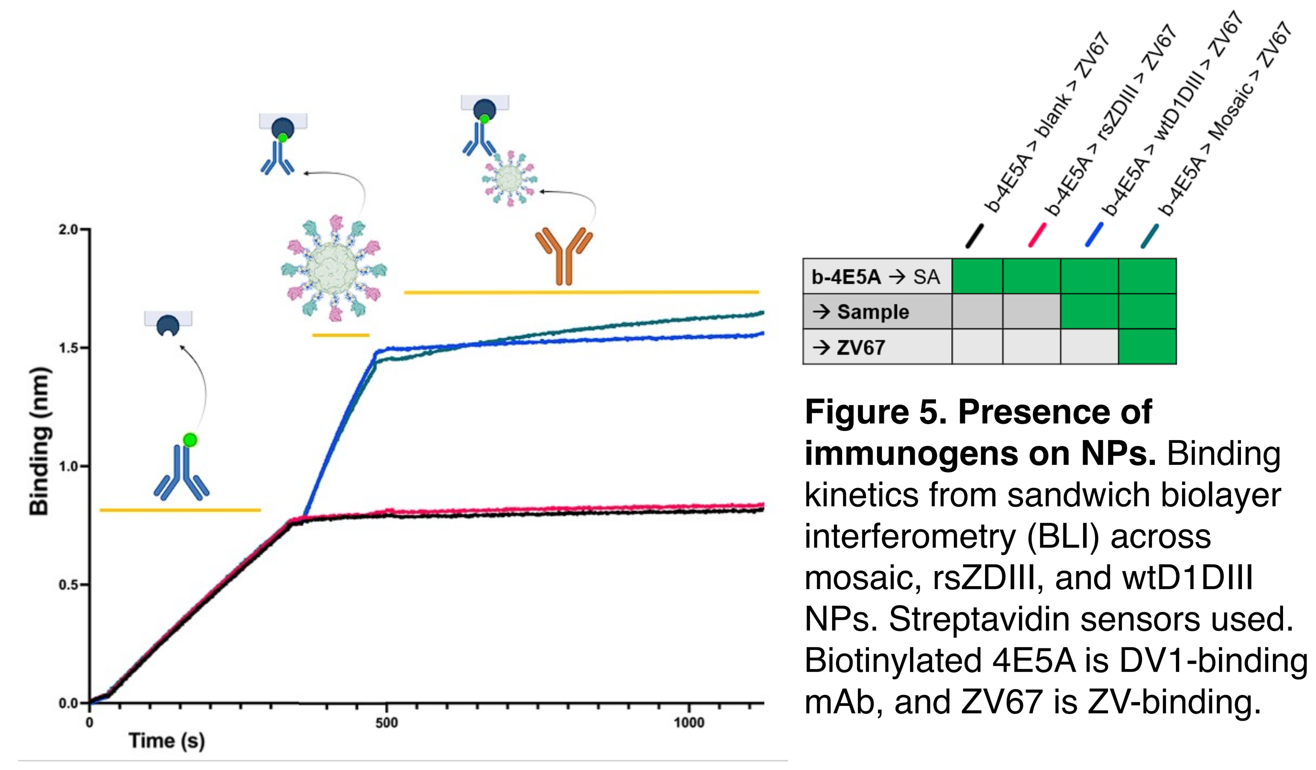


Figure 5. Presence of immunogens on NPs. Binding kinetics from sandwich biolayer interferometry (BLI) across mosaic, rsZDIII, and wtD1DIII NPs. Streptavidin sensors used. Biotinylated 4E5A is DV1-binding mAb, and ZV67 is ZV-binding.

RESULTS (cont'd)

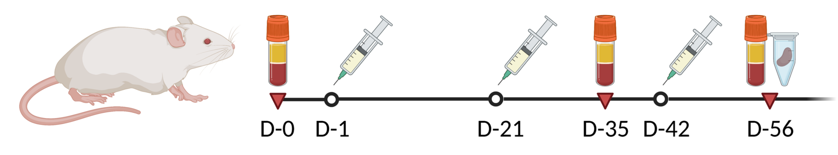


Figure 6. Immunization procedure. 6-week-old BALB/c mice (n = 10). Intraperitoneal Injections were three weeks apart. Submandibular bleeds on Day 0 and Day 35. Cardiac punctures, spleen harvest, and bone collection on Day 56.

- D-56 sera (Fig. 6) across all groups exhibited strong reactivity against all DIIIs (Fig. 7)

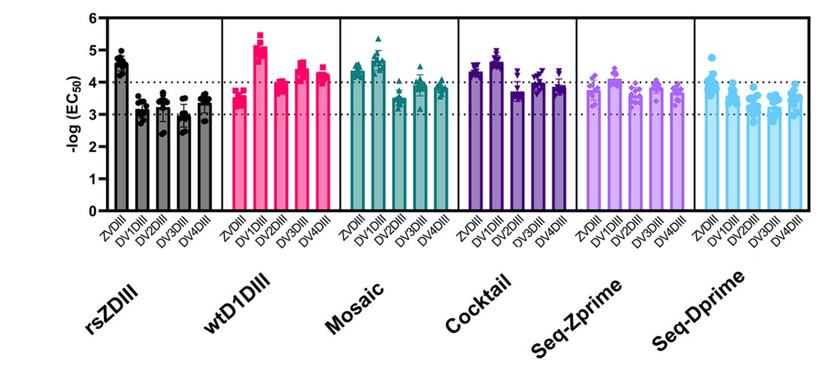


Figure 7. ELISA reactivities of sera. EC50 values, against DIIIs, were compiled by vaccination group. Analysis performed per mouse and repeated twice in technical duplicates.

- Mosaic and Cocktail sera from D-56 demonstrated broad neutralization activity (Fig. 8)

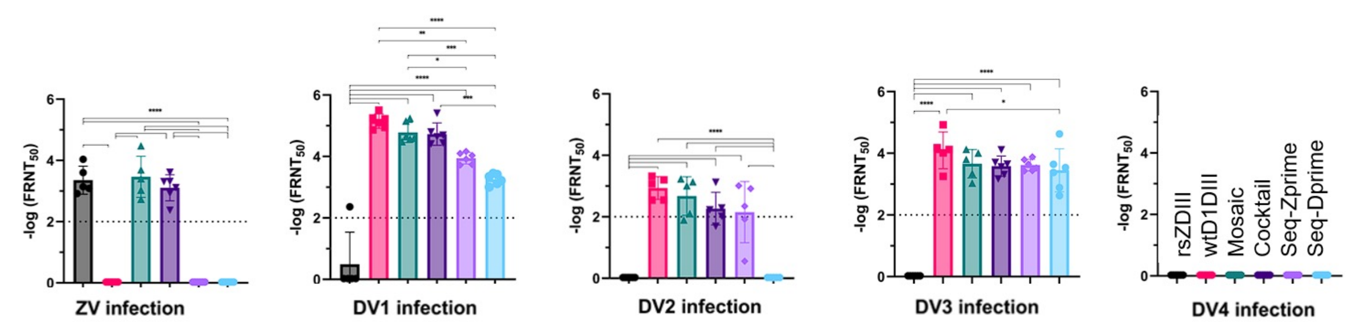


Figure 8. D-56 sera antibody neutralization. Vero cells incubated with sera were subject to infections across ZV and DV1-4. Neutralization activity assessed using FRNT IC_{50}'s. Analysis performed per pooled sera and repeated twice in technical triplicates.

- Preliminary FACS analysis indicated broad humoral response in Mosaic- and Cocktail-immunized mice (Fig. 9)

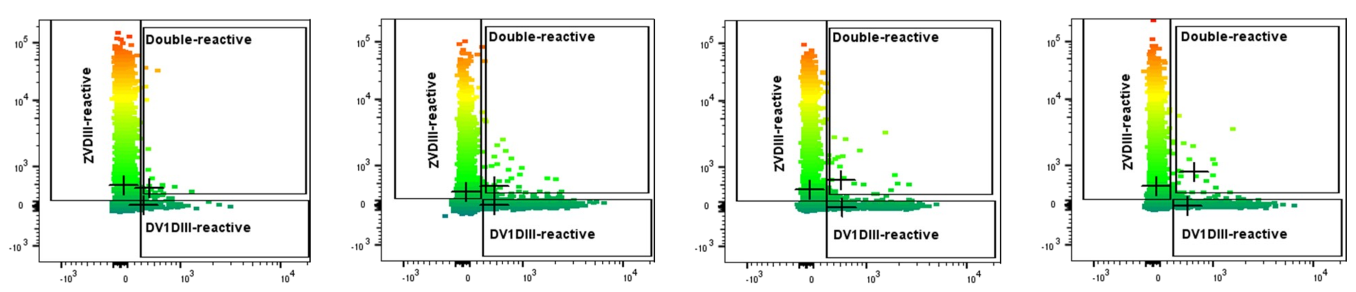


Figure 9. Antigen-positive memory B cells across NP formats. Splenocytes stained for memory B cells that bind to DV1DIII and/or ZVDIII. [L to R] Cell reactivities from *rsZDIII, wtD1DIII, Mosaic,* and *Cocktail* immunization groups. FACS performed per pooled splenocytes and repeated twice. '+' represents median signal of each gate.

FUTURE DIRECTIONS

- Isolate double antigen-positive candidates from Mosaic splenocytes and generate IgG constructs *in vitro*
- Characterize binding, neutralization, and other functional properties of monoclonal antibodies

ACKNOWLEDGEMENTS

This work was supported by R01-125462 & R01-158194 from the National Institute of Allergy and Infectious Diseases and T32-007288 from the National Institute of General Medical Sciences for the Medical Scientist Training Program (MSTP).
I thank Dr. Jonathan R. Lai and my colleagues for their support. I also thank collaborators for their expertise and their contributions.

REFERENCES

1. Georgiev, G.I. *et al*. Resurfaced ZIKV EDIII nanoparticle immunogens elicit neutralizing and protective responses *in vivo*. *Cell. Chem. Bio.* **29**, 1-13 (2022).

Figs. 1, 2, 4, and 6 created with BioRender.